

Market Feasibility & Competitive Positioning Study for ColdEasy

A decentralized cold-storage innovation for rural India — market research conducted under RuTAG / ICAR-NAARM

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Note: This case study presents the research methodology, competitive analysis, and market segmentation work from this engagement. Commercially sensitive details (financials, production capacity, internal operations) belonging to the partner organization have been intentionally omitted out of respect for confidentiality.

1. Project Context

RuTAG (Rural Technology Action Group) is a Government of India initiative that channels engineering and research capacity from premier technical institutions toward grassroots rural technology challenges. As part of an engagement coordinated through ICAR-NAARM, I led primary market research for ColdEasy — a decentralised, dew-point-based cold storage unit developed by Temperate Technologies — to assess its market viability, competitive position, and go-to-market priorities across rural and semi-urban India.

The engagement combined founder-level interviews, structured stakeholder questionnaires, and competitive benchmarking, guided by a research methodology introduced by the project's research mentor.

2. The Research Problem

India's agricultural supply chains lose a significant share of perishable produce every year due to inadequate cold-chain infrastructure, particularly at the farm-gate level where smallholders and FPOs have the least access to storage. ColdEasy was developed to address this gap with a low-power, refrigerant-free cooling unit — but its commercial viability depended on answering three questions: who should it be sold to first, how does it compare to existing alternatives, and where is addressable demand large enough to justify go-to-market investment.

3. Methodology — The Research Framework

Rather than starting with data collection, the project followed a structured Need → Problem → Objective framework:

- Need for research: identify gaps in market knowledge, customer segmentation, and understanding of customer needs and preferences that were limiting go-to-market effectiveness.
- Research problem: defined narrowly and explicitly to keep the research actionable within budget and timeline constraints — geography, sample size, and time horizon were scoped before fieldwork began.
- Research objectives: structured around the classic STP (Segmentation – Targeting – Positioning) model and the 4Ps, applied to a rural, price-sensitive, and largely informal customer base.
- Feasibility was assessed end-to-end across four stages — Production → Distribution → Marketing → After-sales — so recommendations covered the full commercial lifecycle, not just initial sales.

4. Primary Research Conducted

Two complementary research instruments were designed and fielded:

- A founder interview with Temperate Technologies' leadership, to build a detailed technical understanding of the product — its dew-point cooling mechanism (a refrigerant-free alternative to compressor-based cooling), its temperature and humidity control range, and the specific produce categories it serves well (fruit ripening, leafy vegetables, tomatoes) versus categories requiring separate handling (onions and garlic, which need lower relative humidity).
- A structured questionnaire built specifically to explore an unaddressed vertical: mushroom cultivation. Mushroom growers have narrow temperature and humidity tolerances at each growth stage and largely lack access to cold storage today, making this a candidate white-space market. The instrument captured climatic requirements by growth stage, current post-harvest loss patterns, and the storage features that would matter most to this segment.
- A second structured questionnaire built to support direct competitor research — covering target audience profiling, product specifications, brand perception, price sensitivity, and head-to-head comparison against ColdEasy.

5. Competitive Landscape

The most direct competitor identified was Rukart Technologies' "Sabjee Cooler," a passive evaporative-cooling unit targeted at small and marginal farmers. The two products sit at different points on the cost-capacity curve:

Dimension	ColdEasy	Sabjee Cooler
Technology	Dew-point cooling (active)	Evaporative cooling (passive)
Capacity	2,000 kg	50–100 kg
Indicative unit price	~₹80,000 (incl. transport & installation)	~₹50,000
Resource requirement	~4 electricity units/day	15–20 litres water/day, no electricity
Target customer	Large farmers, FPOs, NGOs	Small & marginal farmers, FPOs, NGOs
Founded	2017	2019

This positioning matters strategically: ColdEasy competes less on affordability and more on scale, active climate control, and applicability to high-value, ripening-sensitive produce (dragon fruit, pomegranate, mango) where passive evaporative technology is less effective. Rukart's stronger social media and brand presence was also noted as a relative weakness for ColdEasy worth addressing through content and storytelling, independent of the underlying technology gap.

6. Market Segmentation & Opportunity Mapping

Based on the founder interview and secondary research, several customer segments were identified and evaluated for go-to-market priority:

- High-value, ripening-sensitive crop farmers (dragon fruit, pomegranate, mango) — the strongest near-term fit, given clear willingness to pay for shelf-life extension on already high-margin produce.
- Mushroom cultivators — a genuine white-space opportunity; this segment currently has limited cold-chain access despite narrow climatic tolerances, significant enough to warrant a dedicated research questionnaire.

- FPOs and NGOs — natural anchor customers given their existing rural distribution infrastructure and aggregation role.
- Exploratory B2B channels (hypermarkets, quick-commerce, institutional kitchens) — flagged as longer-term opportunities, with a caveat: many already operate air-conditioned environments, so the value proposition here would need to rest on a different differentiator (e.g., crate-based produce display and overnight storage) rather than raw cooling.

7. Key Recommendations

- Prioritize high-value perishable crop farmers as the primary near-term segment, where unit cost is most easily justified by produce value saved.
- Treat mushroom cultivation as a distinct, underexplored vertical worth a dedicated pilot, given the clarity of the cold-chain gap identified in primary research.
- Invest in brand visibility and founder-led storytelling — a gap identified relative to Rukart, whose stronger digital presence was seen as a credibility advantage despite a comparatively weaker product.
- Approach B2B/institutional channels cautiously and with a sharper value proposition, since “we already have cooling” is a common and valid objection in these segments.

8. Reflection — Skills Applied

This project required structuring an open-ended business problem into a researchable question, designing two distinct primary research instruments from scratch, conducting a founder-level technical interview, building a competitive benchmarking framework, and translating qualitative findings into segmentation and go-to-market recommendations — the core toolkit of market research and business analysis work.